

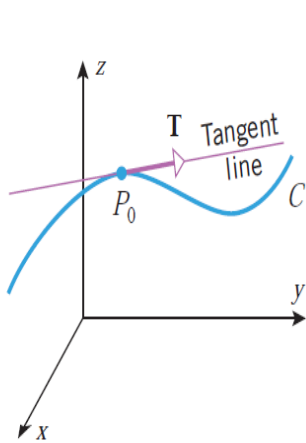
13.7 TANGENT PLANES AND NORMAL VECTORS

In this section we will discuss tangent planes to surfaces in three-dimensional space. We will be concerned with three main questions:

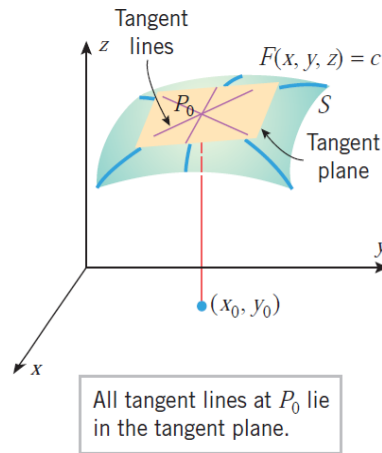
- What is a tangent plane?
- When do tangent planes exist?
- How do we find equations of tangent planes?

13.7.1 DEFINITION Assume that $F(x, y, z)$ has continuous first-order partial derivatives and that $P_0(x_0, y_0, z_0)$ is a point on the level surface $S: F(x, y, z) = c$. If $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$, then $\mathbf{n} = \nabla F(x_0, y_0, z_0)$ is a **normal vector** to S at P_0 and the **tangent plane** to S at P_0 is the plane with equation

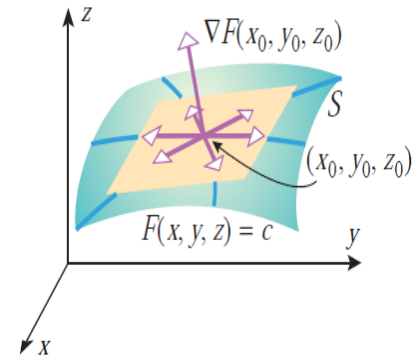
$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0 \quad (3)$$



▲ Figure 13.7.1



▲ Figure 13.7.2



▲ Figure 13.7.3

The line through the point P_0 parallel to the normal vector $\mathbf{n}$ is perpendicular to the tangent plane (3). We will call this the **normal line**, or sometimes more simply the **normal** to the surface $F(x, y, z) = c$ at P_0 . It follows that this line can be expressed parametrically as

$$x = x_0 + F_x(x_0, y_0, z_0)t, \quad y = y_0 + F_y(x_0, y_0, z_0)t, \quad z = z_0 + F_z(x_0, y_0, z_0)t \quad (4)$$

► **Example 1** Consider the ellipsoid $x^2 + 4y^2 + z^2 = 18$.

- (a) Find an equation of the tangent plane to the ellipsoid at the point $(1, 2, 1)$.
- (b) Find parametric equations of the line that is normal to the ellipsoid at the point $(1, 2, 1)$.
- (c) Find the acute angle that the tangent plane at the point $(1, 2, 1)$ makes with the xy -plane.

Solution (a). We apply Definition 13.7.1 with $F(x, y, z) = x^2 + 4y^2 + z^2$ and $(x_0, y_0, z_0) = (1, 2, 1)$. Since

$$\nabla F(x, y, z) = \langle F_x(x, y, z), F_y(x, y, z), F_z(x, y, z) \rangle = \langle 2x, 8y, 2z \rangle$$

we have

$$\mathbf{n} = \nabla F(1, 2, 1) = \langle 2, 16, 2 \rangle$$

Hence, from (3) the equation of the tangent plane is

$$2(x - 1) + 16(y - 2) + 2(z - 1) = 0 \quad \text{or} \quad x + 8y + z = 18$$

Solution (b). Since $\mathbf{n} = \langle 2, 16, 2 \rangle$ at the point $(1, 2, 1)$, it follows from (4) that parametric equations for the normal line to the ellipsoid at the point $(1, 2, 1)$ are

$$x = 1 + 2t, \quad y = 2 + 16t, \quad z = 1 + 2t$$

Solution (c). To find the acute angle θ between the tangent plane and the xy -plane, we will apply Formula (9) of Section 11.6 with $\mathbf{n}_1 = \mathbf{n} = \langle 2, 16, 2 \rangle$ and $\mathbf{n}_2 = \langle 0, 0, 1 \rangle$. This yields

$$\cos \theta = \frac{|\langle 2, 16, 2 \rangle \cdot \langle 0, 0, 1 \rangle|}{\|\langle 2, 16, 2 \rangle\| \|\langle 0, 0, 1 \rangle\|} = \frac{2}{(2\sqrt{66})(1)} = \frac{1}{\sqrt{66}}$$

Thus,

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{66}} \right) \approx 83^\circ$$

■ TANGENT PLANES TO SURFACES OF THE FORM $z = f(x, y)$

To find a tangent plane to a surface of the form $z = f(x, y)$, we can use Equation (3) with the function $F(x, y, z) = z - f(x, y)$.

► **Example 2** Find an equation for the tangent plane and parametric equations for the normal line to the surface $z = x^2y$ at the point $(2, 1, 4)$.

Solution. Let $F(x, y, z) = z - x^2y$. Then $F(x, y, z) = 0$ on the surface, so we can find the gradient of F at the point $(2, 1, 4)$:

$$\nabla F(x, y, z) = -2xy\mathbf{i} - x^2\mathbf{j} + \mathbf{k}$$

$$\nabla F(2, 1, 4) = -4\mathbf{i} - 4\mathbf{j} + \mathbf{k}$$

From (3) the tangent plane has equation

$$-4(x - 2) - 4(y - 1) + 1(z - 4) = 0 \quad \text{or} \quad -4x - 4y + z = -8$$

and the normal line has equations

$$x = 2 - 4t, \quad y = 1 - 4t, \quad z = 4 + t \quad \blacktriangleleft$$

Suppose that $f(x, y)$ is differentiable at a point (x_0, y_0) and that $z_0 = f(x_0, y_0)$. It can be shown that the procedure of Example 2 can be used to find the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0) . This yields an alternative equation for a tangent plane to the graph of a differentiable function.

13.7.2 THEOREM *If $f(x, y)$ is differentiable at the point (x_0, y_0) , then the tangent plane to the surface $z = f(x, y)$ at the point $P_0(x_0, y_0, f(x_0, y_0))$ [or (x_0, y_0)] is the plane*

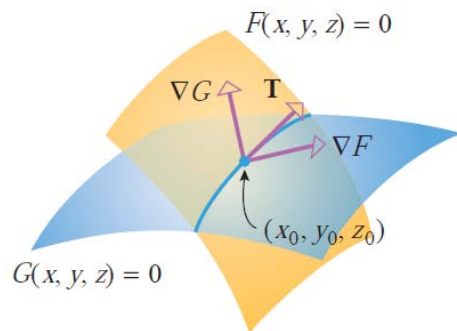
$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \quad (5)$$

■ USING GRADIENTS TO FIND TANGENT LINES TO INTERSECTIONS OF SURFACES

In general, the intersection of two surfaces $F(x, y, z) = 0$ and $G(x, y, z) = 0$ will be a curve in 3-space. If (x_0, y_0, z_0) is a point on this curve, then $\nabla F(x_0, y_0, z_0)$ will be normal to the surface $F(x, y, z) = 0$ at (x_0, y_0, z_0) and $\nabla G(x_0, y_0, z_0)$ will be normal to the surface $G(x, y, z) = 0$ at (x_0, y_0, z_0) . Thus, if the curve of intersection can be smoothly parametrized, then its unit tangent vector $\mathbf{T}$ at (x_0, y_0, z_0) will be orthogonal to both $\nabla F(x_0, y_0, z_0)$ and $\nabla G(x_0, y_0, z_0)$ (Figure 13.7.6). Consequently, if

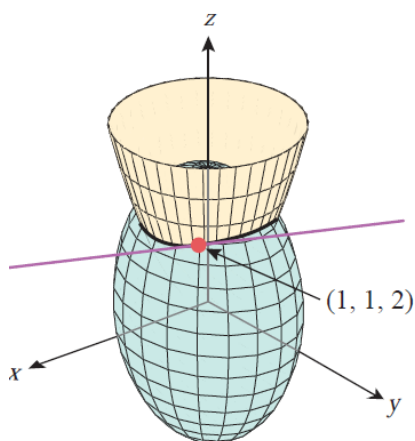
$$\nabla F(x_0, y_0, z_0) \times \nabla G(x_0, y_0, z_0) \neq \mathbf{0}$$

then this cross product will be parallel to $\mathbf{T}$ and hence will be tangent to the curve of intersection. This tangent vector can be used to determine the direction of the tangent line to the curve of intersection at the point (x_0, y_0, z_0) .



▲ Figure 13.7.6

► **Example 3** Find parametric equations of the tangent line to the curve of intersection of the paraboloid $z = x^2 + y^2$ and the ellipsoid $3x^2 + 2y^2 + z^2 = 9$ at the point $(1, 1, 2)$



Solution. We begin by rewriting the equations of the surfaces as

$$x^2 + y^2 - z = 0 \quad \text{and} \quad 3x^2 + 2y^2 + z^2 - 9 = 0$$

and we take

$$F(x, y, z) = x^2 + y^2 - z \quad \text{and} \quad G(x, y, z) = 3x^2 + 2y^2 + z^2 - 9$$

We will need the gradients of these functions at the point $(1, 1, 2)$. The computations are

$$\begin{aligned} \nabla F(x, y, z) &= 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}, & \nabla G(x, y, z) &= 6x\mathbf{i} + 4y\mathbf{j} + 2z\mathbf{k} \\ \nabla F(1, 1, 2) &= 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}, & \nabla G(1, 1, 2) &= 6\mathbf{i} + 4\mathbf{j} + 4\mathbf{k} \end{aligned}$$

Thus, a tangent vector at $(1, 1, 2)$ to the curve of intersection is

$$\nabla F(1, 1, 2) \times \nabla G(1, 1, 2) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & -1 \\ 6 & 4 & 4 \end{vmatrix} = 12\mathbf{i} - 14\mathbf{j} - 4\mathbf{k}$$

Since any scalar multiple of this vector will do just as well, we can multiply by $\frac{1}{2}$ to reduce the size of the coefficients and use the vector of $6\mathbf{i} - 7\mathbf{j} - 2\mathbf{k}$ to determine the direction of the tangent line. This vector and the point $(1, 1, 2)$ yield the parametric equations

$$x = 1 + 6t, \quad y = 1 - 7t, \quad z = 2 - 2t \quad \blacktriangleleft$$

Example 1 Find the equation of the tangent plane to $z = \ln(2x + y)$ at $(-1, 3)$.

Solution

There really isn't too much to do here other than taking a couple of derivatives and doing some quick evaluations.

$$f(x, y) = \ln(2x + y) \qquad z_0 = f(-1, 3) = \ln(1) = 0$$

$$f_x(x, y) = \frac{2}{2x + y} \qquad f_x(-1, 3) = 2$$

$$f_y(x, y) = \frac{1}{2x + y} \qquad f_y(-1, 3) = 1$$

The equation of the plane is then,

$$\begin{aligned} z - 0 &= 2(x + 1) + (1)(y - 3) \\ z &= 2x + y - 1 \end{aligned}$$

One nice use of tangent planes is they give us a way to approximate a surface near a point. As long as we are near to the point (x_0, y_0) then the tangent plane should nearly approximate the function at that point. Because of this we define the **linear approximation** to be,

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

and as long as we are "near" (x_0, y_0) then we should have that,

$$f(x, y) \approx L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Example 2 Find the linear approximation to $z = 3 + \frac{x^2}{16} + \frac{y^2}{9}$ at $(-4, 3)$.

Solution

So, we're really asking for the tangent plane so let's find that.

$$f(x, y) = 3 + \frac{x^2}{16} + \frac{y^2}{9} \qquad f(-4, 3) = 3 + 1 + 1 = 5$$

$$f_x(x, y) = \frac{x}{8} \qquad f_x(-4, 3) = -\frac{1}{2}$$

$$f_y(x, y) = \frac{2y}{9} \qquad f_y(-4, 3) = \frac{2}{3}$$

The tangent plane, or linear approximation, is then,

$$L(x, y) = 5 - \frac{1}{2}(x + 4) + \frac{2}{3}(y - 3)$$

Practice Problem:

Chapter-13.7:

Problems: 1, 2, 3, 5, 10, 11, 12

Thank you for your attention!